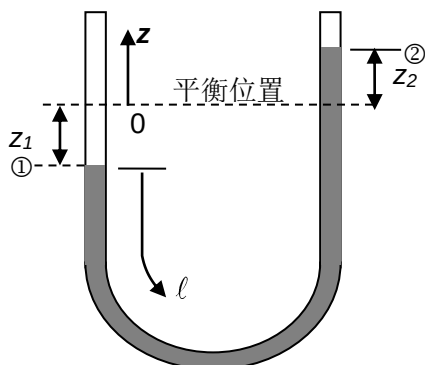


● 应用举例

1. 等截面开口 U 型管内总长 L 的水柱，因初始扰动水柱偏离平衡位置液面差为 h_0 ，开始在管内振荡，不计粘性，求振荡周期，振荡规律和振荡速度。



解：无粘，不可压，一维流动，建立坐标如图，某 t 时刻对两端的液面有

$$\int_1^2 \frac{\partial \vec{V}}{\partial t} \cdot d\vec{l} + \left(\frac{V^2}{2} + \frac{p}{\rho} + gz \right)_2 = \left(\frac{V^2}{2} + \frac{p}{\rho} + gz \right)_1$$

不可压，所以 $V_1 = V_2 = V(t) = \frac{dz_2}{dt}$ ， $p_1 = p_2 = p_a$

$$L \frac{d^2 z_2}{dt^2} + gz_2 = gz_1 \quad z_1 = -z_2$$

$$L \frac{d^2 z_2}{dt^2} + 2gz_2 = 0$$

$$\frac{d^2 z_2}{dt^2} + \frac{2g}{L} z_2 = 0, (r^2 + \frac{2g}{L} = 0, r = \pm i \sqrt{\frac{2g}{L}}) \quad \text{2 阶齐次常系数微分方程求解}$$

$$z_2 = c_1 \cos(\sqrt{\frac{2g}{L}} t) + c_2 \sin(\sqrt{\frac{2g}{L}} t)$$

$$r_1 \neq r_2, y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$$

$$r_1 = r_2, y = (c_1 + c_2 x) e^{rx}$$

$$r = c \pm di, y = e^{cx} (c_1 \cos dx + c_2 \sin dx)$$

$$t = 0: z_2 = \frac{h_0}{2}, V_2 = \frac{dz_2}{dt} = 0$$

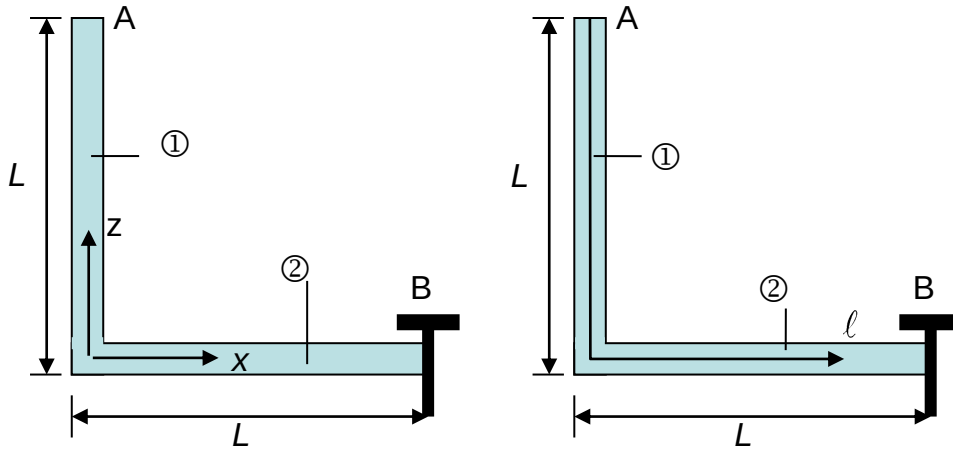
$$\therefore c_1 = \frac{h_0}{2}, c_2 = 0$$

$$z_2 = \frac{h_0}{2} \cos \sqrt{\frac{2g}{L}} t, \quad \text{振荡周期}$$

$$T = \frac{2\pi}{\sqrt{\frac{2g}{L}}} = 2\pi \sqrt{\frac{L}{2g}}$$

$$V_2 = -\frac{h_0}{2} \sqrt{\frac{2g}{L}} \sin(\sqrt{\frac{2g}{L}} t)$$

装满水的等截面直角管放水问题。A 端敞开，B 端接有阀门，在 $t=0$ 时，突然打开阀门使 B 端与大气相通。（1）问 $t=0$ 瞬时，管内各点压力分布？（2） $t>0$ 时，管内各点压力分布，自由面的运动规律。



解：因为不可压，全管每时刻 $V(t)$ 相同，所以 $\frac{dV}{dt} = \text{const}$

$$(1) \quad t=0: \quad A \text{ 到 } B \quad \int_A^B \frac{dV}{dt} d\ell + \frac{p_a}{\rho} + \frac{V_B^2}{2} = \frac{p_a}{\rho} + \frac{V_A^2}{2} + gL$$

$$\frac{dV}{dt} 2L = gL \quad \therefore t=0: \quad \frac{dV}{dt} = \frac{g}{2}$$

采用左图坐标：

$$t=0 \text{ 竖直管内任意一点 } 1: \quad \int_1^B \frac{dV}{dt} d\ell + \frac{p_a}{\rho} + \frac{V_B^2}{2} = \frac{p_1}{\rho} + \frac{V_1^2}{2} + gz_1$$

$$\frac{g}{2}(L+z_1) + \frac{p_a}{\rho} = \frac{p_1}{\rho} + gz_1, \quad \therefore p_1 = p_a + \frac{\rho g}{2}(L-z_1) \quad z_1 > 0$$

$$t=0 \text{ 水平管内任意一点 } 2: \quad \int_2^B \frac{dV}{dt} d\ell + \frac{p_a}{\rho} + \frac{V_B^2}{2} = \frac{p_2}{\rho} + \frac{V_2^2}{2}$$

$$\frac{g}{2}(L-x_2) + \frac{p_a}{\rho} = \frac{p_2}{\rho}, \quad \therefore p_2 = p_a + \frac{\rho g}{2}(L-x_2) \quad x_2 > 0$$

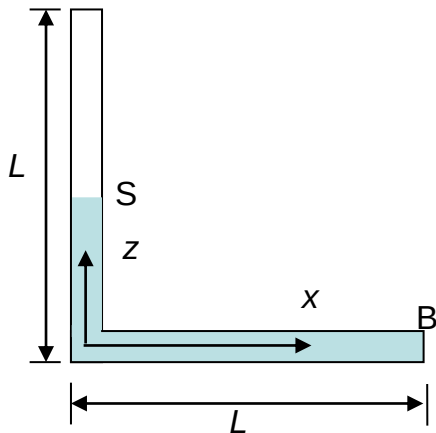
采用右图流线坐标：

$$t=0 \text{ 竖直管内任意一点 } 1: \quad \int_1^B \frac{dV}{dt} d\ell + \frac{p_a}{\rho} + \frac{V_B^2}{2} = \frac{p_1}{\rho} + \frac{V_1^2}{2} + g(L-\ell_1)$$

$$\frac{g}{2}(2L-\ell_1) + \frac{p_a}{\rho} = \frac{p_1}{\rho} + g(L-\ell_1), \quad \therefore p_1 = p_a + \frac{\rho g}{2}\ell \quad 0 \leq \ell \leq L$$

$$t=0 \text{ 水平管内任意一点 } 2: \quad \int_2^B \frac{dV}{dt} d\ell + \frac{p_a}{\rho} + \frac{V_B^2}{2} = \frac{p_2}{\rho} + \frac{V_2^2}{2}$$

$$\frac{g}{2}(2L - \ell_2) + \frac{p_a}{\rho} = \frac{p_2}{\rho}, \quad \boxed{\therefore p_2 = p_a + \rho g(L - \frac{\ell}{2}), \quad L \leq \ell \leq 2L}$$



(2) $t > 0$, (竖直) 自由面到 B 端

$$\int_s^B \frac{dV}{dt} d\ell + \frac{p_a}{\rho} + \frac{V_B^2}{2} = \frac{p_a}{\rho} + \frac{V_s^2}{2} + gh_s$$

$$V_B = V_s = V(t) = -\frac{dh_s}{dt}$$

$$(h_s + L) \frac{dV_s}{dt} = gh_s, \quad \frac{dV_s}{dt} = \frac{h_s}{h_s + L} g, \quad \frac{dV_s}{dh_s} \frac{dh_s}{dt} = \frac{h_s}{h_s + L} g$$

$$-V_s \frac{dV_s}{dh_s} = \frac{h_s}{h_s + L} g, \quad \text{积分可得} \quad \frac{V_s^2}{2} = -g[(L + h_s - L \ln(L + h_s))] + c$$

当 $h_s = L$: $V_s = 0$, 所以 $c = gL(2 - \ln 2L)$

自由面的运动规律为:

$$\frac{V_s^2}{2} = gL(2 - \ln 2L) - g[(L + h_s - L \ln(L + h_s))]$$

$$V_s = \sqrt{2g(L - h_s) + 2gL \ln \frac{L + h_s}{2L}}$$

由 $\frac{dh_s}{dt} = \sqrt{2g(L - h_s) + 2gL \ln \frac{L + h_s}{2L}}$ 可积分出 $h_s(t)$

进一步: 问竖直管中水排放完时的时间, $h_s(t) = 0$, 解出 t

$$t > 0, \text{ 竖直管内任意一点 } \int_1^B \frac{dV}{dt} d\ell + \frac{p_a}{\rho} + \frac{V_B^2}{2} = \frac{p_1}{\rho} + \frac{V_1^2}{2} + gz_1$$

$$\frac{dV_s}{dt} = \frac{h_s}{h_s + L} g,$$

$$\frac{h_s}{h_s + L} g (L + z_1) + \frac{p_a}{\rho} = \frac{p_1}{\rho} + g z_1, \quad \therefore p_1 = p_a + \rho g L \frac{h_s - z_1}{h_s + L}, \quad z_1 > 0$$

$$t=0 \text{ 水平管内任意一点 2: } \int_2^B \frac{dV}{dt} d\ell + \frac{p_a}{\rho} + \frac{V_B^2}{2} = \frac{p_2}{\rho} + \frac{V_2^2}{2}$$

$$\frac{h_s}{h_s + L} g (L - x_2) + \frac{p_a}{\rho} = \frac{p_2}{\rho} \quad \therefore p_2 = p_a + \rho g h_s \frac{L - x_2}{h_s + L} \quad x_2 > 0$$

● 书 4.11

$$\varphi = \frac{Q}{4\pi R} = \frac{-4\pi R_b^2 \dot{R}_b}{4\pi R} = \frac{-R_b^2 \dot{R}_b}{R}, \quad \frac{\partial \varphi}{\partial t} = -\frac{(R_b^2 \ddot{R}_b + 2R_b \dot{R}_b^2)}{R}$$

$$R = R_b \sim R \rightarrow \infty$$

$$-\frac{(R_b^2 \ddot{R}_b + 2R_b \dot{R}_b^2)}{R_b} + \frac{\dot{R}_b^2}{2} + \frac{p_b}{\rho} = \frac{p_\infty}{\rho}, \quad p_b \equiv 0 \quad 10$$

$$\frac{p_\infty}{\rho} + \frac{3\dot{R}_b^2}{2} + R_b \ddot{R}_b = 0$$

$$R_b \ddot{R}_b = R_b \frac{d\dot{R}_b}{dt} = R_b \frac{d\dot{R}_b}{dR_b} \frac{dR_b}{dt} = R_b \dot{R}_b \frac{d\dot{R}_b}{dR_b} = R_b \frac{1}{2} \left(\frac{d\dot{R}_b^2}{dR_b} \right)$$

$$\frac{p_\infty}{\rho} + \frac{3\dot{R}_b^2}{2} + R_b \frac{1}{2} \left(\frac{d\dot{R}_b^2}{dR_b} \right) = 0$$

$$\frac{d\dot{R}_b^2}{\frac{2p_\infty}{\rho} + 3\dot{R}_b^2} = -\frac{dR_b}{R_b}, \quad \ln\left(\frac{2p_\infty}{\rho} + 3\dot{R}_b^2\right) = -3\ln R_b + c$$

$$(\dot{R}_b)_{t=0} = 0, \quad (R_b)_{t=0} = a, \quad c = \ln\left(\frac{2p_\infty}{\rho}\right) + 3\ln a \quad 10$$

$$\dot{R}_b = -\left\{ \frac{2p_\infty}{3\rho} \left[\left(\frac{a}{R_b} \right)^3 - 1 \right] \right\}^{\frac{1}{2}}$$

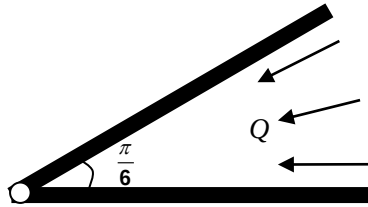
$$t = \sqrt{\frac{3\rho}{2p_\infty}} \int_0^a \frac{R_b^{\frac{3}{2}}}{\sqrt{a^3 - R_b^3}} dR_b$$

● 例 题

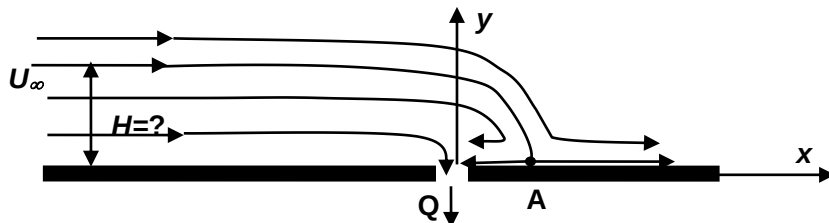
1. 30 度墙角地漏，看做平面点汇，其速度势和速度场

$$\varphi = -\frac{Q}{\frac{\pi}{6}} \ln r = -\frac{6Q}{\pi} \ln r$$

$$V_r = \frac{\partial \varphi}{\partial r} = -\frac{6Q}{\pi r}$$



2. 水以 1.5m/s 均匀流过一平面壁，泵从一缝隙中以 $Q=0.01\text{m}^3/\text{s}$ (单位宽度) 的流量抽吸壁面的水，求驻点位置；求过驻点的流线方程；问距离壁面 $H=?$ 多远的流体不会被吸入缝隙中？



解：均匀流+点汇的例子（解叠加，等流函数为流线，流函数之差为流量）

$$\varphi = U_{\infty}x - \frac{Q}{\pi} \ln \sqrt{x^2 + y^2}, \quad \psi = yU_{\infty} - \frac{Q}{\pi} \arctg \frac{y}{x}$$

$$u = \frac{\partial \varphi}{\partial x} = U_{\infty} - \frac{Q}{2\pi} \left(\frac{2x}{x^2 + y^2} \right), \quad v = \frac{\partial \varphi}{\partial y} = -\frac{Q}{2\pi} \left(\frac{2y}{x^2 + y^2} \right)$$

驻点 $u=v=0$, 驻点 A 位置 $\therefore \begin{cases} y=0 \\ x = \frac{Q}{\pi U_{\infty}} \end{cases} (0, 2.12\text{mm})$

求 H 两种方法：

1. 过驻点 A 的流线在来流方向 $x \rightarrow -\infty$ 的高度为 H , $U_{\infty}H = Q, \therefore H = \frac{Q}{U_{\infty}} = 6.66\text{mm}$

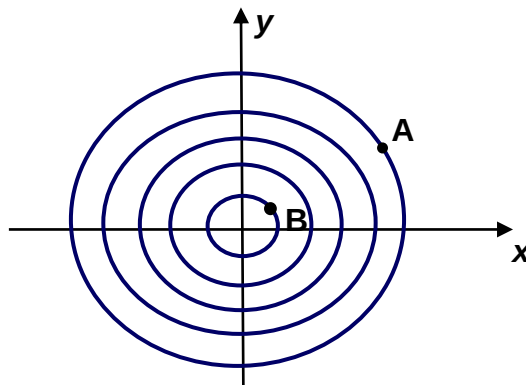
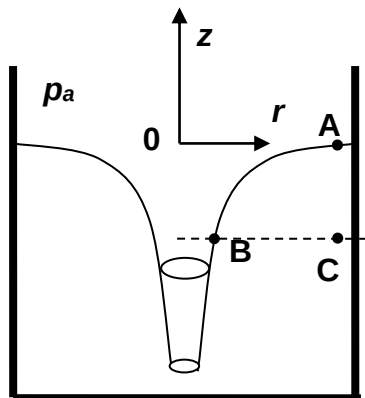
2. 计算驻点 A $(0, \frac{Q}{\pi U_{\infty}})$ 所在的流线上的流函数值 (A 坐标代入流函数)，得到 $\psi = 0$

$$y=0 \text{ 的负 } x \text{ 轴的流线方程为 } \psi = yU_{\infty} - \frac{Q}{\pi} \arctg \frac{y}{x} = -Q$$

流函数之差为流量, $0 - (-Q) = U_{\infty}H, \therefore H = \frac{Q}{U_{\infty}} = 6.66\text{mm}$

● 例题

海底漩涡，在远离中心的速度分布可以近似看成一个平面点涡 $\varphi = \frac{\Gamma}{2\pi} \theta, \psi = -\frac{\Gamma}{2\pi} \ln r$ ，要求确定自由面的形状。



$v_\theta = \frac{1}{r} \frac{\partial \phi}{\partial \theta} = \frac{\Gamma}{2\pi r}$ ，自由面上任选 B 远离小孔，A 处 $r \rightarrow \infty$ ，速度为零，静力学

$$\frac{p_a}{\rho} + \frac{1}{2} v_A^2 = \frac{p_a}{\rho} + \frac{1}{2} v_B^2, \quad \frac{1}{2} v_B^2 + g z_B = 0, \quad z_B = z_s, \quad \frac{1}{2} \frac{\Gamma^2}{4\pi^2 r^2} + g z_s = 0$$

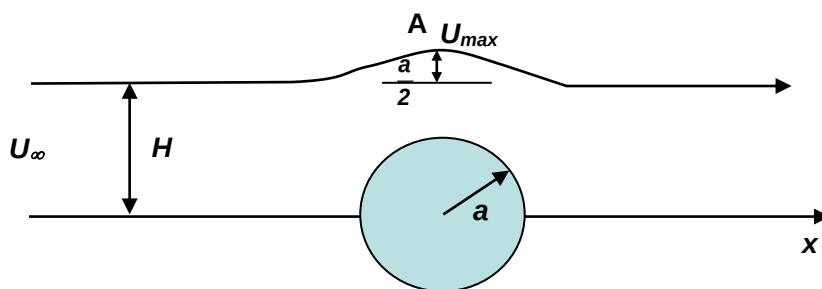
$$\therefore z_s = -\frac{\Gamma^2}{8g\pi^2 r^2},$$

漩涡不能终止在海水中（因为终止处速度会无穷大），所以会一直延伸到海底。（注意：这里不是涡管，是自由表面）

C 点压力 $p_c = p_a + \rho gh$ ，B 点 p_a ，所以涡旋会把海水中的东西吸过来

● 例题

要做一个流线型物体，其外形与无穷远流绕 $r=a$ 圆柱流动的流场中一根流线形状相同，如图所示，问这根流线的相对圆柱对称轴的位置 $H=?$ 其上的最大速度与来流速度之比为多少？



解：可直接用基本解或基本叠加解。

$\psi = U_\infty (r - \frac{a^2}{r}) \sin \theta$ ，对于过 A 点 $\theta = \pi/2$, $r = H + a/2$ 的流线方程，可写为

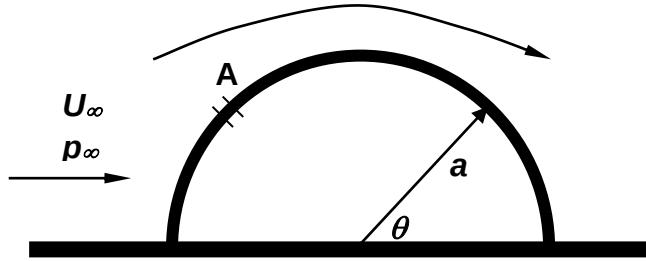
$$\psi = U_\infty (r - \frac{a^2}{r}) \sin \theta = U_\infty (H + \frac{a}{2}) [1 - \frac{a^2}{(H + \frac{a}{2})^2}]$$

X 轴和物面的流线流函数为零

$$\text{两流线之差为之间的流量 } U_\infty (H + \frac{a}{2}) [1 - \frac{a^2}{(H + \frac{a}{2})^2}] - 0 = U_\infty H, \quad \therefore H = \frac{3}{2} a$$

$$\text{A 点 } \theta = \pi/2, r = H + a/2 = 2a, \quad v_r = 0, v_\theta = -U_\infty (1 + \frac{a^2}{4a^2}) = -\frac{5U_\infty}{4}, \quad U_{\max} = -v_\theta = \frac{5U_\infty}{4}, \quad \frac{U_{\max}}{U_\infty} = \frac{5}{4}$$

已知无穷远来流的风速为 U_∞ ，压力为 p_∞ ，绕过一个半径为 a ，长度为 L 的半圆柱型帐篷，设内部压力为 p_i ，用无粘理论推导由于内外压差帐篷受到的升力的表达式。如果风很大，在帐篷表面开个孔 A，使得内压和该处表面压力相等，问 A 孔对应的 $\theta = ?$ 时使得帐篷受到的升力为零。（若是半球如何？）



$$p_{r=a} = p_\infty + \frac{1}{2} \rho U_\infty^2 (1 - 4 \sin^2 \theta)$$

$$F_y = p_i 2a + \iint -p_{r=a} dA \sin \theta = 2p_i a - \int_0^\pi [p_\infty + \frac{1}{2} \rho U_\infty^2 (1 - 4 \sin^2 \theta)] \sin \theta a d\theta$$

$$F_y = 2ap_i - 2ap_\infty + \frac{5}{3} \rho a U_\infty^2$$

若开孔后升力为 0， $p_i = p_\infty + \frac{1}{2} \rho U_\infty^2 (1 - 4 \sin^2 \theta)$ ，代入 $F_y = 0$ ，

$$\text{则 } \rho a U_\infty^2 (1 - 4 \sin^2 \theta) + \frac{5}{3} \rho a U_\infty^2 = 0, \therefore \sin^2 \theta = \frac{2}{3}, \theta = 54.7^\circ, 125.3^\circ$$

● 如果是半球体

$$p_{R=a} = p_\infty + \frac{1}{2} \rho U_\infty^2 (1 - \frac{9}{4} \sin^2 \theta)$$

$$F_y = p_i \pi a^2 + \iint -p_{r=a} \sin \theta \cos \varepsilon dA \quad dA = a^2 \sin \theta d\theta d\varepsilon \quad \theta = 0 \sim \pi, \varepsilon = 2(0 \sim \frac{\pi}{2})$$

$$F_y = \pi a^2 p_i - 2 \int_0^{\frac{\pi}{2}} \int_0^\pi [p_\infty + \frac{1}{2} \rho U_\infty^2 (1 - \frac{9}{4} \sin^2 \theta)] \sin \theta \cos \varepsilon a^2 \sin \theta d\theta d\varepsilon$$

$$= \pi a^2 p_i - \pi a^2 (p_\infty + \frac{1}{2} \rho U_\infty^2) + \rho U_\infty^2 \frac{9}{4} a^2 \int_0^{\frac{\pi}{2}} \cos \varepsilon d\varepsilon \int_0^\pi \sin^4 \theta d\theta$$

$$= \pi a^2 p_i - \pi a^2 (p_\infty + \frac{1}{2} \rho U_\infty^2) + \rho U_\infty^2 \frac{9}{4} a^2 \frac{3\pi}{8}$$

$$= \pi a^2 p_i - \pi a^2 p_\infty + \frac{11}{32} \pi a^2 \rho U_\infty^2$$

若开孔后升力为 0， $p_i = p_\infty + \frac{1}{2} \rho U_\infty^2 (1 - \frac{9}{4} \sin^2 \theta)$ ，代入 $F_y = 0$ ，

$$\text{则 } \frac{\pi a^2 \rho U_\infty^2}{2} (1 - \frac{9}{4} \sin^2 \theta) + \frac{11}{32} \pi a^2 \rho U_\infty^2 = 0, \therefore \sin^2 \theta = \frac{3}{4}, \theta = 60^\circ, 120^\circ$$

在 $\theta = 60^\circ$ 、 120° 开孔升力为 0。

